

Exercise 1a

Set $|1\rangle$ to have 0 energy. Move to rotating frame, valid if $\delta(t)$ assumed to vary slowly compared to ..., apply RWA to get

$$\mathcal{H} = -\hbar\delta(t)|2\rangle\langle 2| - \hbar\left(\frac{\Omega}{2}|2\rangle\langle 1| + \frac{\Omega}{2}|1\rangle\langle 2|\right) \simeq -\hbar\begin{pmatrix} 0 & \Omega/2 \\ \Omega/2 & \delta(t) \end{pmatrix}$$

Eigenvalues λ of matrix given by

$$-\lambda(\delta(t) - \lambda) - \frac{\Omega^2}{4} = 0 \implies E_{\pm} = -\frac{\hbar}{2}(\delta \pm \underbrace{\sqrt{\delta^2 + \Omega^2}}_{:=\Omega_R})$$

with normalized eigenstates

$$\begin{aligned} E_A &= -\frac{\hbar}{2}(\delta + \Omega_R), & |A(t)\rangle &= \Omega|1\rangle - \frac{2}{\hbar}E_B|2\rangle \\ E_B &= -\frac{\hbar}{2}(\delta - \Omega_R), & |B(t)\rangle &= \Omega|1\rangle - \frac{2}{\hbar}E_A|2\rangle \end{aligned}$$

We leave $|A(t)\rangle$ and $|B(t)\rangle$ unnormalized, but denote their square norms by

$$N_A = \langle A(t)|A(t)\rangle = \Omega^2 + \frac{4E_B^2}{\hbar^2}, \quad N_B = \langle B(t)|B(t)\rangle = \Omega^2 + \frac{4E_A^2}{\hbar^2}$$

As $\delta \rightarrow \infty$, then $\Omega_R \rightarrow \delta$ and we get

$$\begin{aligned} E_A &\rightarrow -\hbar\delta, & |A(t)\rangle &\rightarrow |1\rangle \\ E_B &\rightarrow 0, & |B(t)\rangle &\rightarrow |2\rangle \end{aligned}$$

As $\delta \rightarrow -\infty$, then $\Omega_R \rightarrow -\delta$ and

$$\begin{aligned} E_A &\rightarrow 0, & |A(t)\rangle &\rightarrow |2\rangle \\ E_B &\rightarrow -\hbar\delta, & |B(t)\rangle &\rightarrow |1\rangle \end{aligned}$$

Exercise 1b

Remark: $i\hbar\left|\dot{A}(t)\right\rangle \neq H|A(t)\rangle$ because $A(t)$ is an instantaneous eigenvector of H and the time dependence is due only to $\delta(t)$. With $|\psi(t)\rangle = a(t)|A(t)\rangle + b(t)|B(t)\rangle$, we get

$$i\hbar\left|\dot{\psi}(t)\right\rangle = i\hbar\left(\dot{a}(t)|A(t)\rangle + a(t)\left|\dot{A}(t)\right\rangle + \dot{b}(t)|B(t)\rangle + b(t)\left|\dot{B}(t)\right\rangle\right) \quad (1)$$

and

$$H|\psi(t)\rangle = a(t)E_A|A(t)\rangle + b(t)E_B|B(t)\rangle \quad (2)$$

Recall that eigenvectors, v_i of hermitian operator, A , with different eigenvalues, λ_i are orthogonal, as

$$\langle v_i | A | v_j \rangle = \lambda_i \langle v_i | v_j \rangle = \lambda_j \langle v_i | v_j \rangle \implies (\lambda_i - \lambda_j) \langle v_i | v_j \rangle = 0$$

Thus, applying $\langle A(t) |$ and $\langle B(t) |$ to both sides of the hamiltonian equation.

$$\begin{aligned} i\hbar \dot{a} N_A + i\hbar a \langle A | \dot{A} \rangle + i\hbar b \langle A | \dot{B} \rangle &= a E_A N_A \\ i\hbar a \langle B | \dot{A} \rangle + i\hbar \dot{b} N_B + i\hbar b \langle B | \dot{B} \rangle &= b E_B N_B \end{aligned}$$

From 1a, we have that

$$|\dot{A}\rangle = -\frac{2}{\hbar} \dot{E}_b |2\rangle = \dot{\delta} |2\rangle, \quad |\dot{B}\rangle = -\frac{2}{\hbar} \dot{E}_A |2\rangle = \dot{\delta} |2\rangle$$

We have the inner products

$$\begin{aligned} \langle A | \dot{B} \rangle &= \dot{\delta} (\delta - \Omega_R) \\ \langle B | \dot{A} \rangle &= \dot{\delta} (\delta + \Omega_R) \end{aligned}$$

To determine $\langle A | \dot{A} \rangle$, we note that

$$N_A = \langle A | A \rangle \implies \frac{dN_A}{dt} = \langle \dot{A} | A \rangle + \langle A | \dot{A} \rangle = 2 \langle A | \dot{A} \rangle$$

We calculate

$$\frac{dN_A}{dt} = \frac{4}{\hbar^2} 2E_B \dot{E}_B = \frac{4}{\hbar^2} 2E_b \cdot \left(-\frac{\hbar}{2} \dot{\delta}\right) = -\frac{4E_b \dot{\delta}}{\hbar} \implies \langle A | \dot{A} \rangle = -\frac{2E_B \dot{\delta}}{\hbar}$$

Similarly,

$$\langle B | \dot{B} \rangle = -\frac{2E_A \dot{\delta}}{\hbar}$$

the last step holds since $|A\rangle$ is imbedded in a real vector space spanned by $|1\rangle, |2\rangle$.

Plugging in these inner products gives us the EOMs

$$\begin{aligned} i\hbar \dot{a} + i\hbar b \frac{\dot{\delta}(\delta - \Omega_R)}{2\Omega_R \Omega} &= a E_A \\ i\hbar a \frac{\dot{\delta}(\delta + \Omega_R)}{2\Omega_R \Omega} + i\hbar \dot{b} &= b E_B \end{aligned}$$

Simplifying gives us

$$\begin{aligned} \dot{a} &= -a \frac{iE_A}{\hbar} - b \frac{\dot{\delta}(\delta - \Omega_r)}{2\Omega_R \Omega} \\ \dot{b} &= -b \frac{iE_B}{\hbar} - a \frac{\dot{\delta}(\delta + \Omega_R)}{2\Omega_R \Omega} \end{aligned}$$

In the adiabatic limit, $\dot{\delta} \approx 0$ since it changes very slowly, hence we get

$$\begin{aligned}\dot{a} &= -a \frac{iE_A}{\hbar} \\ \dot{b} &= -b \frac{iE_B}{\hbar}\end{aligned}$$

this implies that our state $|\psi(t)\rangle$ just picks up phase factors on the basis states and does not mix states.

Exercise 1c

We have $a(t) = a(0)e^{-iE_A t/\hbar}$ and $b(t) = b(0)e^{-iE_B t/\hbar}$. For $|\psi(0)\rangle = |1\rangle$, we have $a(0) = 1$ and $b(0) = 0$. Thus,

$$|\psi(t)\rangle = e^{-iE_A t/\hbar}|1\rangle$$